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Section : ~~10~~ 10

CSE ~~1000~~ 423

Ans. to the ques. no. 1

Here,

$$\Delta x = 423 - 120 = 303$$

$$\Delta y = 428 - 23 = 405$$

y axis is dominant here due to $|\Delta y| > |\Delta x|$.

$$\therefore \text{Steps} = |\Delta y| = 405$$

$\therefore$ Increment per step -

$$x = \frac{\Delta x}{\text{steps}} = \frac{303}{405} \approx 0.748$$

$$y = \frac{\Delta y}{\text{steps}} = \frac{405}{405} = 1$$

$\therefore x$ will be increased 405 times

$$\begin{aligned}\therefore \text{Total pixels needed} &= \text{steps} + 1 = (405 + 1) \\ &= 406\end{aligned}$$

Ans. to the ques. no. 2

When the slope is between -1 & 1 , the line is more horizontal. Here for every step in x , y changes by 1 unit. So incrementing x by 1 & computing y from the line equation gives a pixel in every column, producing a connected line.

Meanwhile, when slope is less than -1 & greater than 1 , the line is more vertical. Here for every step in x , y changes by more than 1 unit. So incrementing x by 1, we would skip multiple values of y , leaving gaps in the line. Thus we increment y by 1 & compute x from the line equation. This guarantees a pixel in every row, keeping the line unbroken.

Ans. to the ques. no. 7

Vector Display

- 1) Draws an image by directing electron beam only along the actual lines of the picture
- 2) Stores the picture as a set of line drawing commands
- 3) Generally more expensive & complex

Raster Display

- 1) Draws an image by scanning the electron beam across the screen row by row.
- 2) Stores the picture as a matrix of pixel intensity values in a frame buffer.
- 3) Cheaper, simple hardware

A frame buffer is a dedicated area of memory that stores the pixel-by-pixel intensity values for an entire screen image.

The raster display uses a frame buffer

a
Given,

Ans. to the ques. no. 8

(5, 16) to (13, 10)

$$\therefore m = \frac{10 - 16}{13 - 5}$$

$$= -\frac{3}{4} = -0.75$$

Since $-1 < m < 1$,

$$x = x + 1$$

$$y = y + |m|$$

<u>x (+1)</u>	<u>y (-0.75)</u>	<u>y (round)</u>	<u>pixels</u>
5	16	16	(5, 16)
6	15.25	15	(6, 15)
7	14.5	14	(7, 14)
8	13.75 13.25	14	(8, 14)
9	13	13	(9, 13)
10	12.25	12	(10, 12)
11	11.5	12	(11, 12)
12	10.75	11	(12, 11)
13	10	10	(13, 10)

$$b) \quad m = \frac{6-0}{4-0} = \frac{6}{4} = \frac{3}{2} = 1.5$$

Since $-1 < m$,

$$y = y + 1$$

$$x = x + \frac{1}{|m|}$$

row $y(+1)$	$x\left(+\frac{1}{1.5}\right)$	$x(\text{round})$	pixels
0	0	0	(0, 0)
1	0.67	1	(1, 1)
2	1.34	1	(1, 2)
3	2.01	2	(2, 3)
4	2.68	3	(3, 4)
5	3.35	3	(3, 5)
6	4.02	4	(4, 6)

Ans. to the ques. no. 10

10 Given,

$$\frac{x}{7} - \frac{y}{12} = 5$$

If $y = 12 \Rightarrow$

$$\frac{x}{7} - 1 = 5$$

or, $\frac{x}{7} = 6$

or, $x = 42$

$\therefore$ Starting point = $(42, 12)$

If $y = 0 \Rightarrow$

$$\frac{x}{7} - 0 = 5$$

or, $x = 35$

$\therefore$ Ending point = $(35, 0)$

Now,

~~dx~~ $dx = 35 - 42 = -7$

$$dy = 0 - 12 = -12$$

$$\therefore |dy| > |dx|$$

$$\therefore \text{Zone} = 5$$

Noco

$$(42, 12) = (-12, -42)$$

$$(35, 0) = (0, -35)$$

$$dx' = 0 + 12 = 12$$

$$dy' = -35 + 42 = 7$$

$$\begin{aligned} D_{\text{init}} &= 2dy' - dx' \\ &= 2(7) - 12 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \Delta NE &= 2(dy' - dx') \\ &= 2(7 - 12) \\ &= 2(-5) \\ &= -10 \end{aligned}$$

$$\begin{aligned} \Delta E &= 2dy' \\ &= 2(7) \\ &= 14 \end{aligned}$$

<u>x'</u>	<u>y'</u>	<u>d</u>	<u>NE / E</u>	<u>$d(\text{update})$</u>	<u>Corrected</u>		<u>Pixels</u>
					<u>x'</u>	<u>y'</u>	
-12	-42	2	NE	$2 + (-10) = -8$	42	12	(42, 12)
-11	-41	-8	E	$-8 + (14) = 6$	41	11	(41, 11)
-10	-41	6	NE	$6 + (-10) = -4$	41	10	(41, 10)
-9	-40	-4	E	$-4 + 14 = 10$	40	9	(40, 9)
-8	-40	10	NE	$10 + (-10) = 0$	40	8	(40, 8)
-7	-39	0	E	$0 + 14 = 14$	39	7	(39, 7)